A Cost Estimating Relationship for Satellite Constellations

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Abstract

We propose a cost estimating relationship (CER) for satellite constellations. The CER predicts the programme cost from platform mass M and unit count N. The dataset holds 25 systems from 1997 to 2025. Every model input carries a public source. The model writes the programme cost as development cost plus production cost on a learning curve, TOTAL $= g[\,\mathrm{DEV}(M) + \mathrm{PFM}(M)\,N^{\mathrm{LRE}}]$. Leave-one-out cross-validation gives $R^2 = 0.945$. A second regime (New-Space) costs 5 to 21 times less at equal M and N. Unit-cost data does not identify the one-time development cost. Five programmes with disclosed budgets fix the development-to-production split by class and bound DEV(M) to a factor of 1.2. On the traditional regime the CER slightly beats the SMAD USCM8+QuickCost benchmark ($R^2 = 0.88$ versus 0.87); for New-Space the benchmark over-predicts unit cost by a factor near 7. A planetary factor $F = 1.14$ carries the CER to Mars orbit.

Keywords: cost estimating relationship, satellite constellation, parametric cost model, learning curve

1. Introduction

Public unit-cost data covers few navigation and communication satellites. The cost models in SMAD use data up to the early 2000s [1]. This paper uses 25 systems launched from 1997 to 2025. The CER predicts programme cost from platform mass M and unit count N. Section 2 gives the data and its sources. Section 3 gives the earlier method, the new model, and the planetary factor. Section 4 gives the results and the comparison with the SMAD USCM8 and QuickCost models. Section 5 gives the limits.

2. Data

Table 1 lists the 25 systems and the source of each mass, unit count, and unit cost. AVG is the average unit cost. N is the unit count. M is the platform wet mass. All costs use FY2025 euros. USD converts to EUR at 0.9306 [2]. Earlier-year costs escalate to 2025 with US CPI [3]. For most systems the platform mass is 0.79 times the satellite mass [4]; where the manufacturer publishes the platform mass, the paper uses it directly. The traditional regime holds 20 systems. The New-Space regime holds 5 systems. The paper includes a system when a public per-unit cost exists.

Where a source gives a programme total, the paper derives the unit cost as the total divided by N and states the derivation. New-Space unit costs come from analyst estimates and are marked. Table 1 cites the mass, unit count and unit cost of every system separately.

Table 1: Dataset (single source of truth: the reproduction notebook). M is the platform wet mass. Reg.: T = traditional, NS = New-Space. * marks a New-Space analyst estimate. Mass, unit count and unit cost are cited separately.

System	M [kg]	N	AVG [€M]	Reg.
Thales HE-R1000	1000 [5]	5 [6]	106.0 [6]	T
GPS Block III	3065 [7]	10 [7]	268.6 [7]	T
Galileo 2 Generation	1817 [8]	12 [9]	133.5 [9]	T
GPS Block IIF	1290 [10]	12 [10]	179.1 [11]	T
GPS Block IIR	1604 [12]	13 [12]	64.2 [13]	T
Globalstar-3	395 [14]	17 [15]	21.9 [15]	T
O3b MEO	553 [16]	20 [17]	36.1 [18]	T
Globalstar 2nd Gen	553 [19]	24 [20]	48.0 [21]	T
Galileo FOC Overall	577 [22]	34 [23]	49.0 [24, 25]	T
GLONASS-K	739 [26]	5 [26]	110.7 [27]	T
GLONASS-M	1118 [28]	51 [28]	37.5 [27]	T
Iridium NEXT	679 [29]	81 [29]	36.9 [30]	T
Telesat Lightspeed	632 [31]	198 [32]	18.6 [33]	T
QZSS QZS-2/3/4	3239 [34]	7 [35]	218.6 [35]	T
ViaSat-3 F1	5070 [36]	3 [36]	664.4 [37]	T
Inmarsat-6	4321 [38]	2 [39]	385.1 [39]	T
AEHF	7110 [40]	6 [40]	965.0 [40]	T
WGS	4730 [41]	12 [42]	335.0 [41]	T
QZSS QZS-5/6/7	3792 [43]	3 [43]	255.6 [44]	T
O3b mPOWER	1343 [45]	8 [46]	158.0 [47]	T
OneWeb Gen-1	116 [48]	648 [49]	0.9 [50]*	NS
Amazon Kuiper	451 [51]	3232 [52]	1.9 [53]*	NS
Starlink v1.5	243 [54]	2987 [55]	0.4 [56]*	NS
Starlink v2-mini	577 [57]	7676 [55]	0.8 [58]*	NS

System	M [kg]	N	AVG [€M]	Reg.
AST BlueBird B2	4819 [59]	60 [60]	19.0 [61]*	NS

3. Method

3.1. Earlier method

The earlier method extracted a per-satellite production cost (PFM) from AVG, N, and M. It set the share $\lambda = \text{PFM}/(\text{PFM} + \text{DEV})$ as a linear function of N, from 0.35 at $N = 5$ to 0.15 at $N = 198$. It then fit $\text{PFM} = 0.149\, M^{0.89}$. This method has two limits. First, the extraction removes N; leave-one-out cross-validation of that CER gives $R^2 = 0.62$. Second, the function $\lambda(N)$ comes from a choice, not from data.

3.2. New model

The new model writes the programme cost as

$$\text{TOTAL}(M, N) = F\, g \left[\text{DEV}(M) + \text{PFM}(M)\, N^{\text{LRE}} \right], \tag{1}$$

with $\text{DEV}(M) = d\, M^p$, $\text{PFM}(M) = q\, M^r$, and $\text{LRE} = 1 + \ln(0.9)/\ln 2 = 0.848$. The learning rate is 0.9 [62]. The term g equals 1 for the traditional regime and 10^{off} for New-Space. The calibrated parameter off fits to -0.92 in log-space. F is the planetary factor (Section 3.3). The programme cost is development plus production: $\text{TOTAL} = \text{DEV} + \text{PROD}$, with $\text{PROD} = \text{PFM}\, N^{\text{LRE}}$. The average unit cost is $\text{AVG} = \text{TOTAL}/N$, so the identity $\text{TOTAL} = \text{AVG} \cdot N$ holds by construction.

Unit-cost data alone does not identify the one-time development cost DEV. A free fit of d and p gives a development band that runs to zero. The paper fixes this with disclosed budgets. Five programmes report development and production cost separately (Table 2). Their development fraction $\varphi = \text{DEV}/\text{TOTAL}$ is a ratio, so it constrains the split without leaving the dataset's cost basis. The five parameters d, p, q, r, off are fit in one least-squares step against two sets of observations: the unit cost AVG of every system, and the disclosed φ of the four bespoke programmes. This keeps the cross-validated fit and bounds $\text{DEV}(M)$ (Section 4.2). The resolved CER is $\text{DEV}(M) = 5.99\, M^{0.613}$ and $\text{PFM}(M) = 0.219\, M^{0.828}$, with $g = 0.120$ for New-Space.

3.3. Planetary cost factor

The dataset holds Earth-orbit satellites. A Mars-orbit constellation costs more. The paper estimates the increase bottom-up. It scales each bus subsystem by a literature cost-increase factor and by the subsystem's share of bus cost [63, 64]. The sum is a $10.0\% \pm 4.5\%$ increase in bus cost. The paper adopts $F = 1.14$ for Mars and $F = 1$ for Earth. The SMAD QuickCost model applies a single 1.29 planetary factor [65]; the subsystem estimate is lower and traceable.

3.4. Validation

Cross-validation rotates the held-out data. Leave-one-out removes one system, refits, and predicts the removed system. Repeated 5-fold removes a set of systems. Bootstrap resampling gives coefficient intervals. The regime is a design input.

4. Results

4.1. Fit and cross-validation

Leave-one-out gives $R^2 = 0.945$ over the 25 systems and $R^2 = 0.845$ on the traditional regime alone. The 1-sigma prediction band is a factor of 1.55. The typical absolute error is 44%. Figure 1 shows predicted against actual cost.

4.2. Development split

For a 1 tonne system, the bootstrap gives DEV = €414 M with a 1-sigma factor of 1.2; the free fit was unbounded, running to zero. Development cost is 57% to 63% of total programme cost for the four bespoke programmes and 11% for the one commercial programme (WGS). Figure 2 shows the two CERs with their bootstrap bands and the disclosed data points each split in production and development.

4.3. New-Space regime

The New-Space regime costs 5 to 21 times less than the traditional regime at equal M and N. The point estimate is 8. The 5 New-Space systems carry masses from 116 to 4819 kg and unit counts from 60 to 7676.

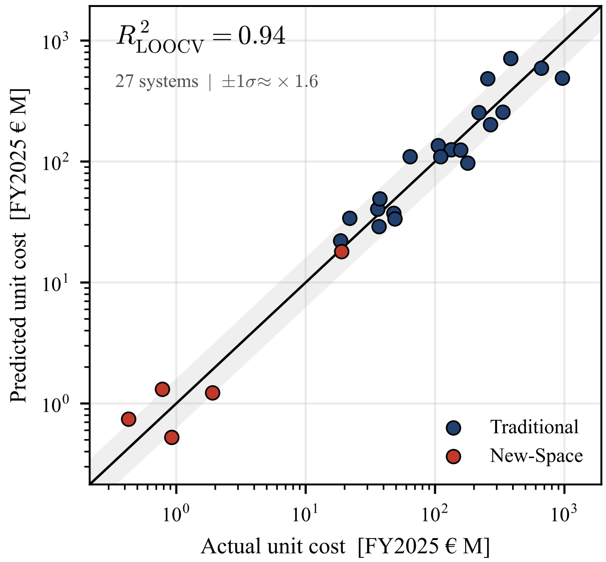

Figure 1: Predicted against actual unit cost. 25 systems. Leave-one-out $R^2 = 0.945$. Band: 1-sigma, factor 1.55.

Table 2: Disclosed development and production cost, as reported in the cited source (US programmes: then-year US$M, SAR current estimate; Galileo: €M). $\varphi = \text{DEV}/(\text{DEV} + \text{PROD})$ is the development fraction and the only anchor quantity the fit uses; it is independent of currency and inflation basis. SAR: Selected Acquisition Report.

Programme	Development	Production	φ	Class	Source
AEHF	$7465.9	$5695.2	0.57	bespoke	[66]
MUOS	$4140.1	$2932.3	0.59	bespoke	[67]
SBIRS	$10304.8	$7176.9	0.59	bespoke	[68]
Galileo	€950.0	€566.0	0.63	bespoke	[69]
WGS	$409.6	$3392.3	0.11	commercial	[70]

4.4. Comparison with USCM8 and QuickCost

The paper compares the CER with the SMAD USCM8 development and production CERs and the QuickCost payload term [71, 72, 65]. On the traditional regime the CER is slightly more accurate: $R^2 = 0.88$ versus 0.87, and a median error of 24% versus 31%. The USCM8 CERs carry a standard error of 47% for development and 21% for production. On the New-Space regime USCM8+QuickCost over-predicts unit cost by a factor near 7, because it has no term for mass production. Figure 3 shows programme cost over M and N for the three streams. The traditional CER tracks the

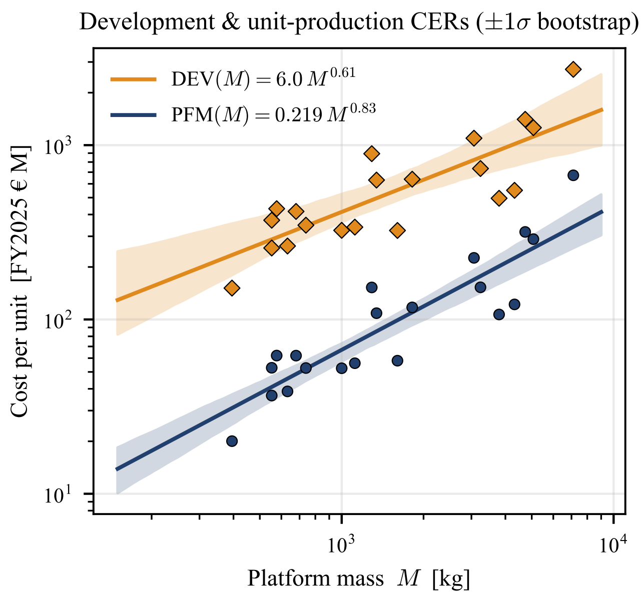

Figure 2: DEV(M) and PFM(M) CER components (lines) with 1-sigma bootstrap bands. Markers: each traditional system's observed programme cost AVG $\cdot$ N split into development (diamonds) and unit production (circles) by the fitted class fraction. The development level is set by the disclosed fractions (Table 2); its band is a factor of 1.2.

benchmark; the New-Space regime sits a full order of magnitude below it.

4.5. Operations

The paper carries the SMAD mission operations model for the operations phase [73]. For a 31-satellite constellation over 5 years it gives €159 M. The drivers are flight and ground software maintenance, 19 engineers and 4 technicians, and facilities.

5. Limitations

The dataset holds 25 systems. The New-Space regime holds 5 systems and is weakly cross-validated. New-Space unit costs come from analyst estimates. The development split depends on programme class. Four of the five anchor programmes are bespoke; the commercial fraction rests on one anchor (WGS). Some descriptive attributes (power, spare count) have no public source and are omitted. The escalation to 2025 uses US CPI.

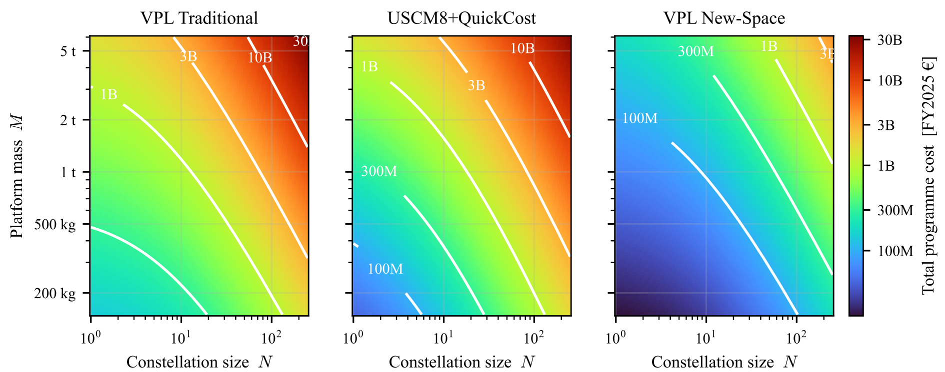

Figure 3: Total programme cost over platform mass and constellation size, with iso-cost contours, for the VPL Traditional CER, the SMAD USCM8+QuickCost benchmark, and the VPL New-Space regime.

6. Conclusion

The CER predicts constellation programme cost from mass and unit count. Leave-one-out gives $R^2 = 0.945$. A second regime costs 5 to 21 times less. Disclosed budgets fix the development-to-production split and bound the development cost to a factor of 1.2. The CER matches the SMAD USCM8+QuickCost benchmark on traditional systems and corrects its over-prediction on New-Space systems.

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